# MICRO LAMBDA WIRELESS, INC.

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## YIG-Tuned Synthesizer

**MODEL NO.** MLVS-2009AS

**Date / Revision** Date 17January2019 Rev -

[X] RoHS Compliant

**Customer** Pennsylvania State University

**Base model** MLVS-0520AS

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Output Frequency Range | 0.5 - 20.0 | GHz | | | | | | |
| Power Output | +15 | dBm | X | min | | typ | | max |
| Step Size | 0.001 | Hz | X | min | | typ | | max |
| Switching Speed (Full Band step) | 50 | uS | | min | | typ | X | max |
| Settling time | +/- 1 | ppm | | min | | typ | X | max |
| Output Impedance | 50 | Ohms | | min | X | typ | | max |
| Load VSWR | 2.0:1 | | | min | X | typ | | max |
| Harmonics (>250MHz) | -12 | dBc | | min | | typ | X | max |
| Harmonics (<250MHz) | -10 | dBc | | min | | typ | X | max |
| Sub-Harmonics | -50 | dBc | | min | | typ | X | max |
| Spurious | -60 | dBc | | min | | typ | X | max |
| Phase Noise | | | | | | | | |
| @ 100 Hz offset | -86 | dBc/Hz | | min | | typ | X | max |
| @ 1 kHz offset | -108 | dBc/Hz | | min | | typ | X | max |
| @ 10 kHz offset | -115 | dBc/Hz | | min | | typ | X | max |
| @ 100 kHz offset | -117 | dBc/Hz | | min | | typ | X | max |
| @ 1 MHz offset | -115 | dBc/Hz | | min | | typ | X | max |
| @ 10 MHz offset | -127 | dBc/Hz | | min | | typ | X | max |
| @ 50 MHz offset | -150 | dBc/Hz | | min | | typ | X | max |
| @ 100 MHz offset | -153 | dBc/Hz | | min | | typ | X | max |

## Reference Oscillator Options

[X] **Internal Reference**

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Output Frequency | 10 | MHz | | min | X | typ | | max |
| Output Power | +5, +/-1dB | dBm | | min | | typ | X | max |
| Stability | +/-1 | ppm | | min | | typ | X | max |
| Aging | +/-0.5 | ppm / year | | min | | typ | X | max |
| Tuning range | +/-2 | ppm | | min | X | typ | | max |
| Output Impedance | 50 | Ohms | | min | X | typ | | max |
| Warm up time | 2 | minutes | | min | X | typ | | max |

[X] **External Reference**

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Input Frequency | 10 | MHz | | min | X | typ | | max |
| Stability for lock | +/-5 | ppm | X | min | | typ | | max |
| Input Level | 0, +/-6dB | dBm | X | min | | typ | X | max |
| Input Impedance | 50 | Ohms | | min | X | typ | | max |

## Supply Voltage & Current

| | |
|---|---|
| +10 Vdc to +16Vdc | 1200 mA @ 12 Vdc Nominal |
| Power Consumption | 15 Watts nominal |

## Digital Control Format

Control Commands

| | |
|---|---|
| DC Connector-20 pin | SPI plus Busy / SCPI / Binary |
| USB connector | USB-HID windows |
| Programming Information | see Manual |

## Screening

| | | |
|---|---|---|
| [ ] | N/A | |
| [X] | MLW Screening | Procedure per Process Flow Diagram |
| [ ] | Customer Screening | Procedure # _____ Revision ___ |
| [ ] | Military | |

## Testing

| | | |
|---|---|---|
| [ ] | Level 1 - No Data | |
| [ ] | Level 2 - Sample Data | Per Lot of _____ |
| [ ] | Level 3 - Summary Data | |
| [X] | Level 4 - Standard Test Data | |
| [ ] | Level 5 - Special Test Requirements | |

## Marking

| | | | | | |
|---|---|---|---|---|---|
| [X] | Standard Labeling | MDL, S/N, D/C, Warranty, ESD if Req'd | | | |
| [ ] | Special Labeling | Customer Part Number | [ ] | [ ] | Customer Cage Code |
| | | Revision | [ ] | [ ] | Micro Lambda Cage Code |
| [ ] | Other | | | | |
| [X] | RoHS | | | | |

## Environmental

| | | | | | |
|---|---|---|---|---|---|
| [X] | Operating Temperature (case) | 0 | °C to | +60 | °C |
| | Storage Temperature | -40 | °C to | +85 | °C |
| | Relative Humidity | 10% to 90%, non-condinsing | | | |
| | Shock | 30g, half-sine pulse, 11msec duration | | | |
| | Vibration | 5Hz to 500Hz, 0.31g rms | | | |

## Case

Outline 99-0101-001 Drawing # 21 oz

**Notes:** This part is equivalent to MLVS-0520AS.